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Course Code: CS2406 **Name of the Programme: B.Tech** **Name of the Course: Incident Response and Digital Forensics** **Semester: IV**		
Course credit	**No. of hours per week (L + T + P)**	**Total no. of Teaching hours**
1:1:2	15+15+30	60
Course Objectives:		
a) Understand the principles of incident response, forensic fundamentals, and legal considerations involved in digital investigations		
b) Learn methods for collecting and preserving network and host-based evidence, including volatile and non-volatile data.		
c) Explore forensic imaging techniques and gain skills in analyzing network, memory, and storage-based digital evidence.		
d) Develop the ability to document forensic findings through formal reports and apply threat intelligence for proactive cyber defense.		

Course outline (Syllabus of the course)

Syllabus:

Module – 1	**No. of hours**
	03
Understanding Incident Response The Incident Response Process, The Incident Response Framework, The Incident Response Plan, Incident Response Playbooks, Testing the Incident Response Framework, Engaging the Incident Response Team, Incorporating crisis communications.	

Module - 2	**No. of hours**
	03
Fundamentals of Digital Forensics and Investigation Methodology Forensic science overview, Trace evidence and digital forensics, Legal issues in digital forensics, Forensic procedures in incident response, AI in digital forensics. Incident investigation analysis types, Functional Digital Forensic Investigation methodology, Cyber Kill Chain, Diamond Model of Intrusion analysis	

Module – 3	**No. of hours**
	03
Collecting Network Evidence and Acquiring Host -Based Evidence, Network evidence overview, Firewalls and proxy logs, NetFlow, Packet capture, Wireshark, Evidence collection. Preparation, Order of volatility, Evidence acquisition, Acquiring volatile memory, Acquiring non-volatile evidence	

Module – 4	**No. of hours**
	03
Analyzing System Memory and Analyzing System Storage Analyzing System Memory, Memory evidence overview, Memory analysis. Analyzing System Storage, Forensic platforms	

Module - 5	**No. of hours**
	03
Forensic Reporting and Threat Intelligence Forensic Reporting, Documentation overview, Incident tracking, Written reports, Threat Intelligence, Threat intelligence overview, Threat intelligence methodology, Threat intelligence direction, Threat intelligence sources, Threat intelligence platforms	

Tutorials: **15 hours**

1. Simulating a Phishing Incident and Incident Response Scenario: Detect and respond to a phishing attack targeting employee emails. Learn how to identify phishing, contain the threat, and communicate with stakeholders.
2. Creating an Incident Response Plan for a Small Business Scenario: Draft a complete IR plan tailored for a company with 50 employees. Focus on roles, escalation paths, and crisis communication.
3. Building Playbooks for Malware and Ransomware Attacks
   Scenario: Develop stepwise playbooks for isolating infected machines and recovering data.
4. Conducting Tabletop Exercises for Incident Response Testing
   Scenario: Run a mock ransomware attack drill with a cross-functional team.
5. Using Ticketing Systems for Incident Tracking and Documentation

Scenario: Log, update, and close incident tickets in a simulated SOC environment.

6. Writing a Forensic Report After a Data Breach

   Scenario: Document evidence from a simulated breach, creating timelines and impact summaries.
7. Log Analysis for Incident Timeline Reconstruction

   Scenario: Analyze system and network logs to recreate the sequence of an intrusion.
8. Collecting Threat Intelligence from Open Sources

   Scenario: Use AlienVault OTX and VirusTotal to gather intelligence on an emerging malware campaign.
9. Mapping Threat Data to MITRE ATT&CK Framework

   Scenario: Analyze attacker behaviors using collected data and classify them per ATT&CK tactics.
10. Creating Threat Intelligence Reports for SOC Analysts

    Scenario: Compile threat data into actionable intelligence reports for security teams.
11. Integrating Threat Intelligence Feeds into SIEM Alerts

    Scenario: Configure SIEM to automatically generate alerts from threat feeds.
12. Threat Hunting Using IOC Data Scenario: Use Indicators of Compromise (IOCs) to search network data for hidden threats

Laboratory Experiment List

List of Experiments
1. Incident Response Simulation Using TheHive: Simulate a malware incident and manage the response lifecycle.   2. Draft an Incident Response Plan Document: Write an IR plan document with templates and roles for a fictional company.

3. Develop an Incident Response Playbook for DDoS Attacks: Create detailed playbook steps to mitigate and respond to a DDoS attack.
4. Tabletop Exercise for Social Engineering Attack Response: Conduct a role-playing exercise responding to a social engineering scenario.
5. Incident Ticket Management in Jira or RT: Create, track, and close tickets related to simulated security events.
6. Write a Forensic Report Based on Simulated Attack Artifacts: Analyze provided logs, extract evidence, and write a formal forensic report.
7. Reconstruct an Attack Timeline Using ELK Stack: Ingest logs into ELK and visualize the sequence of an intrusion.
8. Collect and Analyze Threat Intelligence from OTX and VirusTotal: Extract relevant threat data and analyze malicious IPs/domains.
9. Use MITRE ATT&CK Navigator to Classify Threats: Map collected indicators to specific ATT&CK tactics and techniques.
10. Prepare a Threat Intelligence Report for SOC Operations: Use real threat data to prepare and present a threat report.
11. Configure Open Source SIEM (Security Onion) to Use Threat Feeds: Integrate a threat intelligence feed and validate generated alerts.
12. Threat Hunt with Zeek (Bro) IDS Using IOC Indicators: Search network traffic captures for known malicious indicators.

Course outcomes
a) Demonstrate the core concepts of incident response, digital forensics, and related legal considerations.
b) Apply standard techniques for collecting and preserving digital evidence from networks and host systems.
c) Analyze forensic images, memory dumps, and network data to extract relevant information.
d) Prepare structured forensic reports and incorporate threat intelligence into incident response.

Reference books	
1	G. Johansen, Digital Forensics and Incident Response: A Practical Guide to Deploying Digital Forensic Techniques in Response to Cyber Security Incidents, 1st ed. Birmingham, U.K.: Packt Publishing, 2017. ISBN: 1787288684
2	2 G. Johansen, Digital Forensics and Incident Response, 4th ed. Birmingham, U.K.: Packt Publishing, 2025. ISBN: 9781836200109.

Lesson Plan

Name of the Programme:	**B.Tech(H)**	
Title of the Course:	**Incident Response and Digital Forensics**	
Course code	**CS2406**	
Semester:	**IV**	
Names of the Course Instructor(s)	**Dr Basavaraj Patil**	
Course credit	**No. of hours per week (L + T + P)**	**Total no. of Teaching hours**
1:1:2	15+15+30	60

Session	Module No. / Tutorial / Lab	Topic / Tutorial / Lab	Pedagogy / Activities	Pre-reading / Reference
1	M1 (L)	Introduction to Incident Response	Interactive lecture with breach examples	Johansen (2017), Ch.1
2	T1	Phishing Incident Identification	Discussion on phishing indicators	CERT-IN advisories
3	Lab	Introduction to Lab	Demonstration	
4	Lab 1	TheHive – Case Creation	Hands-on IR case setup	TheHive Docs
5	M1 (L)	Incident Response Lifecycle	Diagram-based explanation	Johansen (2025), Ch.2
6	T1	Phishing Response & Containment	Scenario walkthrough	Johansen (2017)
7	Lab	Malware IR Simulation in TheHive	Case handling	TheHive Docs
8	Lab	Malware IR Simulation in TheHive	Case handling & task assignment	TheHive Docs
9	M1 (L)	IR Frameworks & Standards	Case comparison (NIST-based)	Johansen (2025), Ch.3
10	T2	IR Plan – Roles & Escalation	Group drafting activity	IR Plan templates

11	Lab 2	Draft IR Plan Document	Hands-on document creation	Johansen (2017)
12	Lab 2	Draft IR Plan Document	Hands-on document creation	Johansen (2017)
13	M1 (L)	IR Playbooks & Crisis Communication	Lecture + discussion	Johansen (2017), Ch.4
14	T3	Malware & Ransomware Playbooks	Playbook design activity	Playbook samples
15	Lab 3	DDoS IR Playbook Development	Stepwise playbook creation	ENISA guidelines
16	Lab 3	DDoS IR Playbook Development	Stepwise playbook creation	ENISA guidelines
17	M1 (L)	Testing IR Frameworks	Lecture on tabletop testing	Johansen (2025), Ch.5
18	T4	Tabletop IR Exercise	Role-based discussion	Instructor notes
19	Lab 4	Tabletop Social Engineering Drill	Role-play execution	
20	Lab 4	Tabletop Social Engineering Drill	Role-play execution	
21	M2 (L)	Digital Forensics Overview	Concept lecture	Johansen (2017), Ch.6
22	T5	Incident Ticketing Concepts	SOC workflow discussion	SOC SOPs
23	Lab 5	Ticket Creation in Jira / RT	Hands-on ticket creation	Jira / RT Docs
24	Lab 5	Ticket Creation in Jira / RT	Hands-on ticket creation	Jira / RT Docs
25	M2 (L)	Legal Issues in Digital Forensics	Case-based lecture	Johansen (2025), Ch.6
26	T5	Incident Ticket Escalation	Ticket lifecycle exercise	SOC templates
27	Lab 5	Ticket Tracking & Closure	Hands-on ticket management	Jira / RT Docs
28	Lab 5	Ticket Tracking & Closure	Hands-on ticket management	Jira / RT Docs
29	M2 (L)	Forensic Procedures in IR	Workflow explanation	Johansen (2017), Ch.7
30	T6	Forensic Report Structure	Report outline discussion	Report templates
31	Lab 6	Forensic Report Writing	Evidence-based report writing	Johansen (2017), Ch.16
32	Lab 6	Forensic Report Writing	Evidence-based report writing	Johansen (2017), Ch.16
33	M3 (L)	Network Evidence Overview	Lecture with examples	Johansen (2017), Ch.10

34	T7	Log Analysis for Timeline	Guided analysis activity	Sample logs
35	Lab 7	ELK – Log Ingestion	Hands-on ELK setup	ELK Docs
36	Lab 7	ELK – Log Ingestion	Hands-on ELK setup	ELK Docs
37	M3 (L)	Packet Capture & Volatility	Lecture + demo	Johansen (2017), Ch.11
38	T7	Timeline Reconstruction	Correlation exercise	Timeline worksheet
39	Lab 7	ELK Timeline Visualization	Dashboard creation	c
40	Lab 7	ELK – Log Ingestion	Hands-on ELK setup	ELK Docs
41	M4 (L)	Memory Forensics Concepts	Concept lecture	Johansen (2017), Ch.13
42	T12	IOC-based Threat Hunting	Guided hunting discussion	SANS IOC refs
43	Lab 12	Threat Hunting with Zeek	Hands-on IOC search	Zeek Docs
44	Lab 12	Threat Hunting with Zeek	Hands-on IOC search	Zeek Docs
45	M4 (L)	Storage & Disk Forensics	Lecture	Johansen (2025), Ch.14
46	T8	Threat Intelligence Sources	OTX & VirusTotal demo	OTX Docs
47	Lab 8	Threat Intelligence Collection	Hands-on TI extraction	VirusTotal Docs
48	Lab 8	Threat Intelligence Collection	Hands-on TI extraction	VirusTotal Docs
49	M5 (L)	Forensic Reporting Standards	Lecture	Johansen (2017), Ch.17
50	T9	MITRE ATT&CK Mapping	ATT&CK classification activity	MITRE ATT&CK
51	Lab 9	ATT&CK Navigator Mapping	Hands-on mapping	MITRE Navigator
52	Lab 9	ATT&CK Navigator Mapping	Hands-on mapping	MITRE Navigator
53	M5 (L)	Threat Intelligence Lifecycle	Lecture	Johansen (2025), Ch.18
54	T10	TI Report for SOC	Report drafting activity	TI templates
55	Lab 10	Threat Intelligence Report	Hands-on report preparation	SOC report samples
56	Lab 10	Threat Intelligence Report	Hands-on report preparation	SOC report samples
57	M5 (L)	Integrating TI with IR	Lecture	Johansen (2025), Ch.19

58	T11	SIEM Alerting from TI Feeds	Configuration discussion	SIEM Docs
59	Lab 11	Security Onion – TI Feed Integration	Hands-on alert validation	Security Onion Docs
60	Lab 12	Threat Hunting with Zeek (Bro) IDS Using IOC Indicators	Simulation	

Note: 1

2 credit courses	**30 hours**
3 credit courses	**45 hours**
4 credit courses	**75 hours**

Note: 2

- **A course shall have a minimum of 3 modules and maximum 5 modules**
- **Each module will be covered in minimum 6 hours and maximum 12 hours**

Assessment and Evaluation

Continuous Internal Evaluation (CIE)	
Components of CIE	Weightage of each component
CIE-1 - Quiz - Assignment	 10 10
CIE-2 - Theory (pen and paper)	25
CIE-3 - Tutorial and Lab report - Case-study Assignment and Presentation	 10 15

Semester End Exams (SEE)	
Mode of Exam	Practical
Weightage of exam	30%

Course Outcomes- Program Outcomes Matrix

Sl.No	**Course outcome**	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	
1	Demonstrate the core concepts of incident response, digital forensics, and related legal considerations.	3	3		2								
2	Apply standard techniques for collecting and preserving digital evidence from networks and host systems.		2	3	2	2							
3	Analyze forensic images, memory dumps, and network data to extract relevant information.			3	2	1			2				
4	Prepare structured forensic reports and incorporate threat intelligence into incident response.	3	2			3							

Rubrics for evaluation of CIE (separate for each CIE component)

CIE Components		Marks	Rubrics
CIE-1	MCQ/MSQ	10	Multiple selection questions of 10 marks
	Assignment	10	Assignment will be evaluated for 10 marks based on assessment criteria prepared by the course faculty.
CIE-2	Pen and Paper	25	Test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty.
CIE-3	Lab report	10	Tutorial and Lab execution reports need to be submitted in soft binding.
	Case-study Assignment & presentation	15	Prepare the research article related to the course and submit the draft copy and presentation.

CIE 1: Assignment (10 Marks)					
Sl. No.	**Criteria**	**Measuring Methods**	**Excellent (8–10)**	**Good (5–7)**	**Poor (0–4)**
1	Case Understanding & Scenario Analysis	Accuracy in identifying the incident, stakeholders, and forensic challenges	Demonstrates precise understandin g of the case, identifies all key issues logically	Shows reasonable understandin g with minor gaps	Limited understanding: key elements misunderstood or missing
2	Forensic Methodology & Process Application	Description of acquisition, preservation, analysis, documentation steps	Applies correct forensic procedures with strong justification and sequence	Applies appropriate methods but with limited justification	Incorrect, incomplete, or missing forensic steps
3	Quality of Evidence Interpretation & Findings	Accuracy of findings, linkage to evidence, clarity in reasoning	Interprets evidence accurately with clear, logical conclusions	Provides partially correct interpretation with some gaps	Incorrect, unclear, or unsupported interpretation of evidence
4	Presentation, Structure & Clarity	Report structure, organization,	Well-structure d, clear, well presented	Adequately structured; minor issues in clarity	Poorly structured, unclear, lacks

		clarity, formatting			professional format

CIE 3- Tutorial and Lab report					
Sl. No.	**Criteria**	**Measuring Methods**	**Excellent (8–10)**	**Good (5–7)**	**Poor (0–4)**
1	Execution of Experiment	Accuracy and completion of lab activity	Fully executed with correct results	Completed with minor errors	Incomplete or wrong results
2	Report Submission	Quality of report (Aim, procedure, results)	Clear, complete, well-structured	Adequate but missing minor details	Poorly documented or incomplete
3	Analysis & Interpretation	Quality of analysis and understanding of results	Insightful interpretation	Limited analysis	No analysis or wrong inference
4	Viva-voce	Q&A	Answer all queries with minor mistakes	Satisfactory	Poor

CIE 3: Case-study (15 Marks)					
Sl. No.	**Criteria**	**Measuring Methods**	**Excellent (8–10)**	**Good (5–7)**	**Poor (0–4)**
1	Problem Identification & Relevance	Clarity and originality of research problem	Innovative and well-defined topic	Relevant but limited scope	Unclear or repetitive topic
2	Content & Research Depth	Quality of literature, methodology, and findings	Comprehensive and well-supported	Adequate but lacks rigor	Poorly supported or incomplete
3	Execution	Accuracy and completion of execution	Fully executed with correct results	Completed with minor errors	Incomplete or wrong results
4	Presentation & Communication	Delivery, visuals, and ability to answer	Clear, confident, and engaging	Clear but lacks depth	Unclear, unprepared